

Technology #06: Neuromorphic Shunt Failure Predictor

Technology Domain: Machine Learning + Clinical Data

Mechanism: ML predictor using ICP + flow trends to forecast shunt failure 24h before symptoms

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: No real failure dataset exists to train the model (CRITICAL BLOCKER per Article XXXVII)

Next Experiment: P-13 ML predictor: Synthetic training dataset (clearly labeled)

Buyer Receives: ML predictor architecture (gradient boosting, 0 prior-art hits); Feature extraction specification (rolling 24h window); Clinical alert threshold ($P > 0.7$); Power dependency analysis (conditional on P-15-R1/P-16); This engineering dossier with data-pipeline-specific design chain

Buyer Must Build: Data partnership for 100+ patient-years training data; Production ML pipeline (training, validation, deployment); SaMD regulatory submission (510(k) or De Novo); Cybersecurity design (if connected device); Clinical validation evidence

Transfer Posture: Sponsored Validation

Target Buyer: Medical device AI/digital health company; neuroshunt manufacturer with digital platform

CONFIDENTIAL. See full engineering dossier for details.